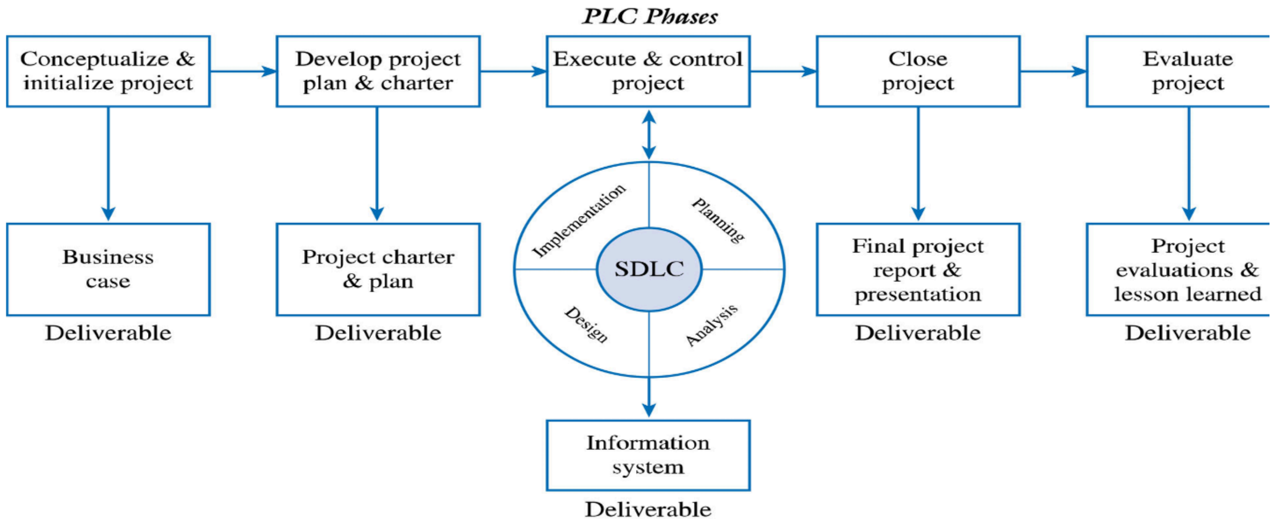
Overview

- Software development lifecycle (SDLC)
    - 5 stages: Planning, Analysis, Design, Implementation, Maintenance

SDLC model	Status of req.	Type of IS	Example
Waterfall	must be clear	mission critical, zero tolerance for mistakes	飛安系統
Prototyping	better be clear	UI, UX centric	遊戲
Spiral	unclear / risk-driven	mass-produced	套裝軟體
Incremental	better be clear	large, customized	客製ERP
Agile	unclear is OK	UI, UX centric	任何軟體都可以使用，但須考慮 communication cost & entire time frame

- ITPM methodology
    - 4 PM Objectives: PM iron triangle

- Project Liftcycle (PLC)



  - 5 phases: Initialization > Planning > Execution > Closure > Evaluation
  - SDLC happened at Execution phase
- 4 lessons learned
  - reusable code
  - best practices
  - new risks & how the team dealt
  - actual data
- 5 PM Processes
  - Initiating > Planning > Executing > Monitoring&Controlling > Closing
  - happened each phase in PLC

Project Initialization

- To define measurable organizational value (MOV)
  - should be measurable, verifiable, provide value, agreed
  - 6 steps:
    1. Impact areas: customer / strategic / financial / operational / social
    2. Value: better / faster / cheaper / do more
    3. Metric
    4. Time: when the MOV be achieved?
    5. Verify: use MVPA
    6. Summarize

       Increase 1000 new HL newsletter subscribers within 2 years or

Time Period	MOV
6 months	250 new healthy living newsletter subscribers
1 year	600 new healthy living newsletter subscribers
2 years	1,000 new healthy living newsletter subscribers

- To establish business case (BC)
  - a proposal that states the reason for an IS investment
  - 四大關鍵要素
    - MOV
    - Alts to achieve MOV
    - Cost-Benefit Analysis
    - Perceived risks
  - 7 steps:
    1. Define MOV
    2. Seek support from BC stakeholders
    3. Identify alternatives (default + 5)
       - do nothing
       - 不使用 IT 來改善流程
       - 採用同業系統
       - 租用市面上的系統
       - 改造現有系統
       - 客製新系統
    4. Identify feasibilities
    5. Define cost and benifits
       - tangible or intangible?
       - one-time or recurring?
    6. Analyze alternatives using financial models and scoring models
    7. Propose and support the recommendation
  - Financial models
    - Breakeven Point
      - $\text{Breakeven Point} = \text{Initial Investment}/\text{Net Profit Margin}$
      - 某系統開發成本 100,000 元，單品獲利 5 元 → 20,000

- Cost-Benefit Analysis

XXXX 企業購置 XXX BI 系統的投資報酬分析

	Year of project							
	year 0	year 1	year 2	year 3	year 4	yaer 5	TOTALS	
Net Economic benefit	$0	$40,000	$40,000	$40,000	$40,000	$40,000		
Discount rate(12%)	1	0.8929	0.7972	0.7118	0.6355	0.5674		
PV of benefits	$0	$35,716	$31,888	$28,472	$25,420	$22,696		
NPV of all BENEFITS	$0	$35,716	$67,604	$96,076	$121,496	$144,192	**$144,192**	總流入
One-Time COSTS	-$60,000							
Recurring Costs	$0	-$6,000	-$6,000	-$6,000	-$6,000	-$6,000		
Discount rate(12%)	1	0.8929	0.7972	0.7118	0.6355	0.5674		
PV of Recurring Costs	$0	-$5,357	-$4,783	-$4,271	-$3,813	-$3,404		
NPV of all COSTS	-$60,000	-$65,357	-$70,141	-$74,411	-$78,224	-$81,629	**-$81,629**	總流出
總淨現值							$62,563	
總投資報酬						ROI=	0.7664354	
損益分析：								
Yearly NPV Cash Flow	-$60,000	$30,359	$27,105	$24,201	$21,607	$19,292		
Overall NPV Cash Flow	-$60,000	-$29,641	-$2,537	$21,665	$43,272	$62,563		
Project break-even occurs between year 2 and 3								
Use first year of positive cash flow to calculate break-even ratio				0.1048				
Actual break-even occurred at 2.1 years								

Note: 1.All dollar values have been rounded to the nearest dollar

2.year 5's discount rate = 1/(1.12)^5=0.5674

- $\text{Net Present Value} = -I_0 + \sum\left(\text{Net Cash Flow}/(1+r)^t\right)$
  - I: total cost
  - r: discount rate
  - t: time period
- $\text{ROI} = \text{NPV}/I$

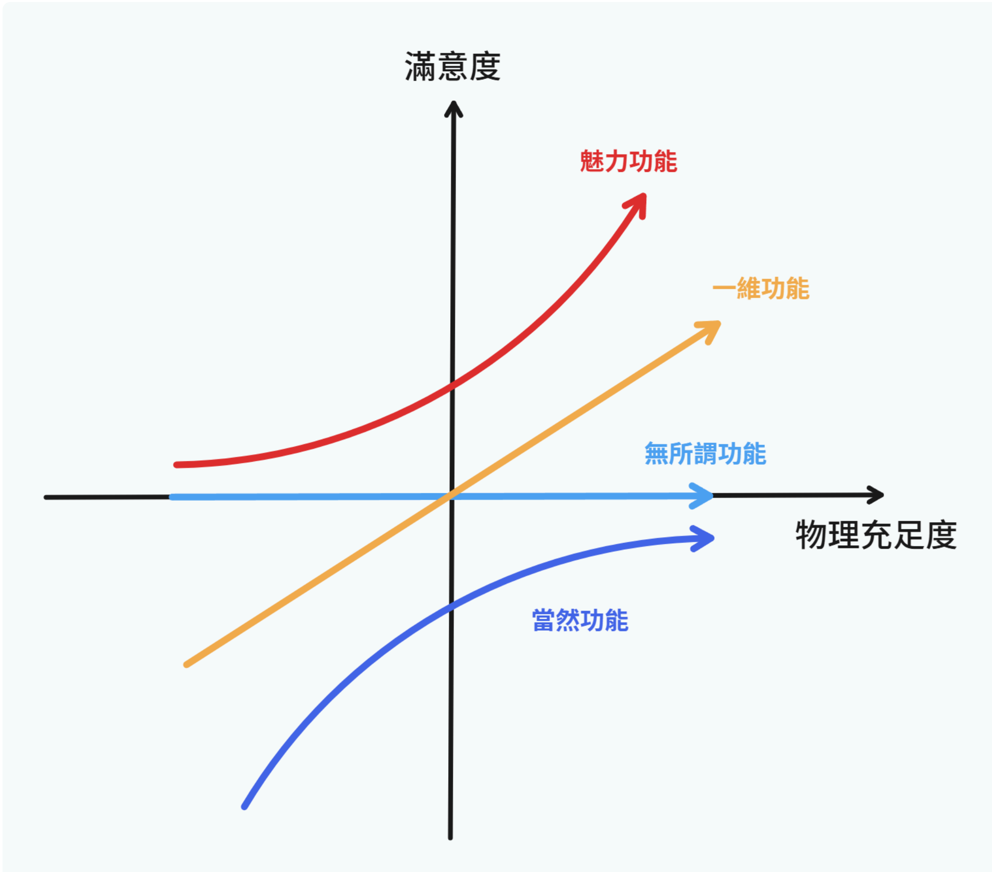
Project Planning

- after BC approved, establish Project Infrastructure
  - Project Charter: 給所有 stakeholders 的原則
  - Project Plan: 給內部 stakeholders 的細節
- to close the planning phase need 3 things:
  - PC + PP + kick-off meeting

Project Scope Mgt.

1. Planning scope
   - Scope is largely determined by requirements:
     - Functional requirement: to make the system run
     - Non-Functional requirement: to make the system run better
2. Collect requirements

1. Gathering: interview, survey, field study, workshops, JAD...
2. Prioritizing:
    - MoSCoW: must, should, could, want
    - The Kano Method:



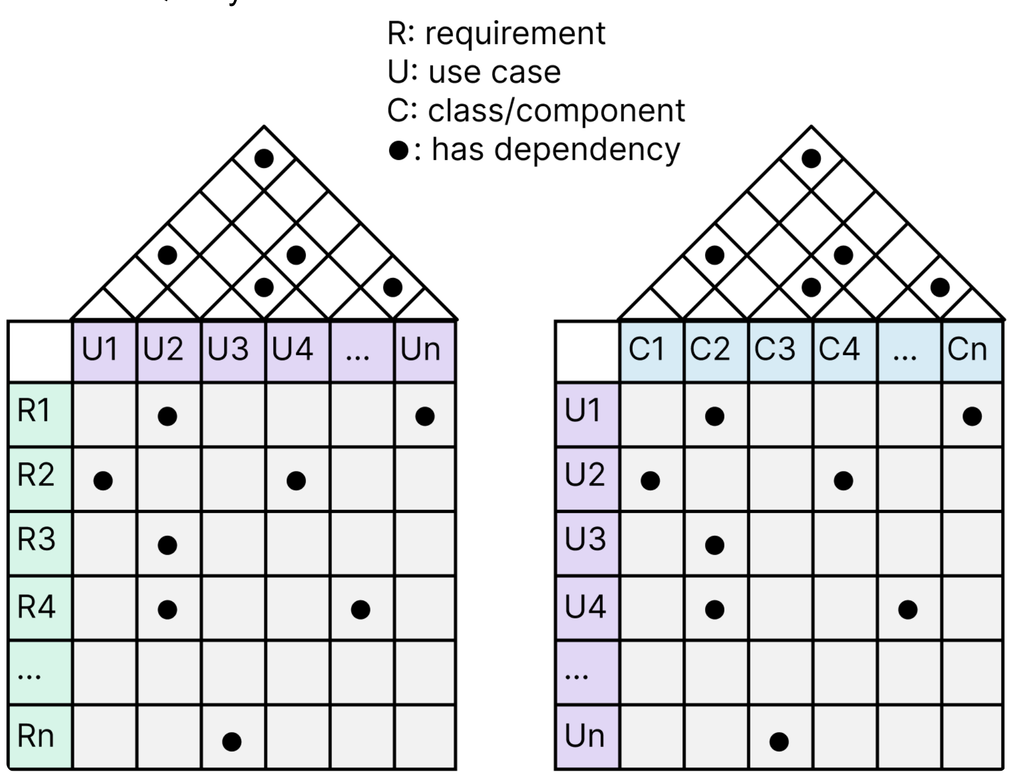
缺少當然會失敗、一維需要維持、魅力是獲利關鍵

3. Defining (the 7 criterias)
    - correct: 符合所需
    - unambiguous: 定義明確
    - verifiable
    - consistent: 跟其他需求不衝突
    - complete: 描述需有使用者與動詞
    - traceable
    - modifiable

3. Defining Scope
    - scope statement: what is and what is not, to be delivered in the project?
    - product scope: what features need in the product?
        - using use-case diagram + context diagram
    - project scope: what deliverables need to support the development
        - using Deliverable Structure Chart
4. Creating WBS

- a tool to define tasks to fulfill the deliverables
- *Tasks* are the lowest level, each < 80 hrs
- *Work package* is a group of tasks
- we can define *Milestones* at the end of work package
- three types:
    - procedure oriented: focus on process (suit for Waterfall)
    - product oriented: focus on output (suit for Agile)
    - hybrid

5. Verifying Scope
6. Controlling Scope
    - mitigates the 4 typical scope mgt. problems:
        - Scope Grope (定義不明確)
        - Scope Creep (增減原本不在範圍內的功能)
        - Scope Leap (範圍巨變)
        - Gold Plating (幫不必要的功能鍍金)
    - tools
        - Requirement Traceability Matrix (RTM):
            - House of Quality

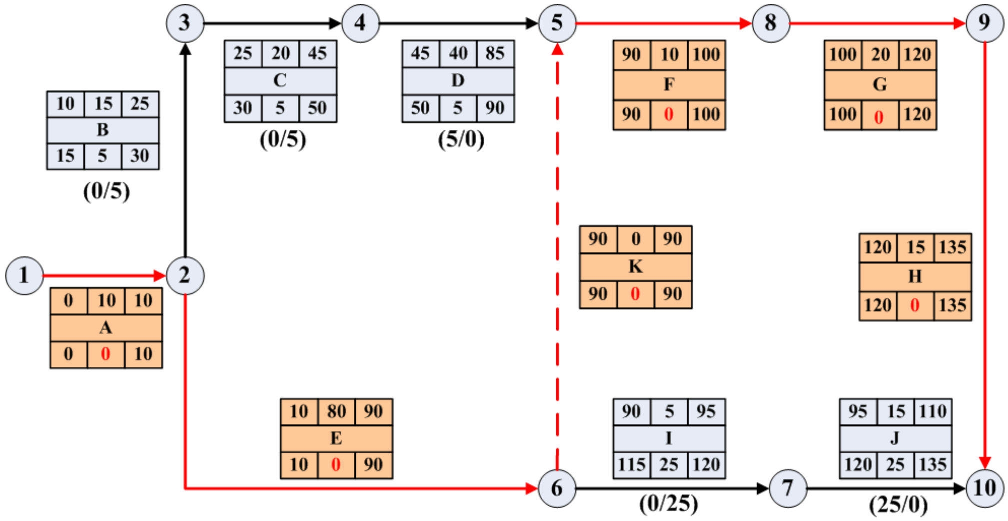


        - Scope change request form

Project Time Mgt.

1. Defining tasks using activity list
2. Sequencing tasks
    - 4 relationships:
        - Finish to Start (FS): 大隊接力
        - Finish to Finish (FF): 最後的 task 結束後專案才能結束
        - Start to Start (SS): 開始寫 testcase 之後測試才可以開始
        - Start to Finish (SF): 老師開始講課後助教才能離開
    - 4 dependencies:
        - Mandatory (強制)
        - Discretionary (不強制)
        - Extrenal (between project and non-project activities)
        - Internal (between project activities)
3. Estimating task resources
    - 3 major resources: People, Equipment, Material
    - consider technical difficulty & availability
4. Estimating task duration
    - Top-down: *sponsor* determine overall time, then breakdown with WBS
    - Bottom-up: *task-owners* determine task time, then sum up
    - Delphi Method: involves experience experts
    - 3-Point Estimate:
        - $\text{Estimate Time} = (o + 4r + p)/6$
            - o: optimistic completion time
            - p: pessimistic completion time
            - r: most likely completion time
            - obtain by previous project's actual data
5. Developing project schedule
    - establish the paths/network of tasks using PDM (Activity on Node) or ADM (Activity on Arrow)

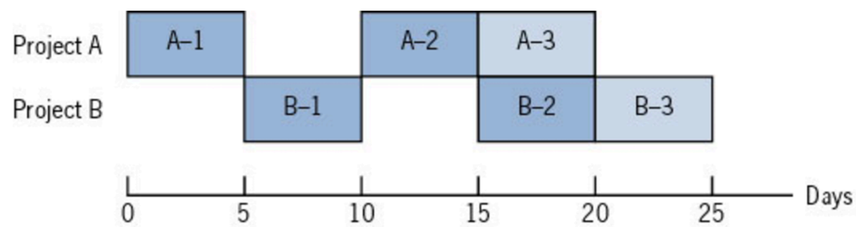
- evaluation using PERT



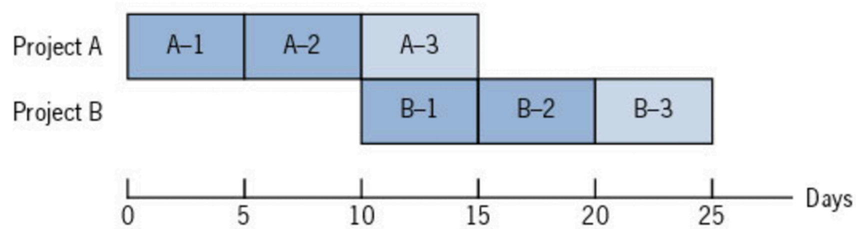
  - for per task: ES, LS, EF, LF
  - slack time = $LS - EF$ = $LF - EF$
  - Critical Path: the bottleneck path with 0 slack time
- PERT 5 steps:
  1. establish WBS to obtain tasks
  2. determine the sequences of the tasks
  3. use 3-point estimation to compute duration
  4. establish project's PERT
  5. eastblish a Gnatt chart

6. Controlling the schedule
   - Time-Boxing
     - fixed time allocated to a planned task, stop working once time up
     - often used in Agile
   - Resource Leveling
     e.g. A1, A2, B1, B2 shared same resources

Solution I: Compromise the schedule by splitting the tasks

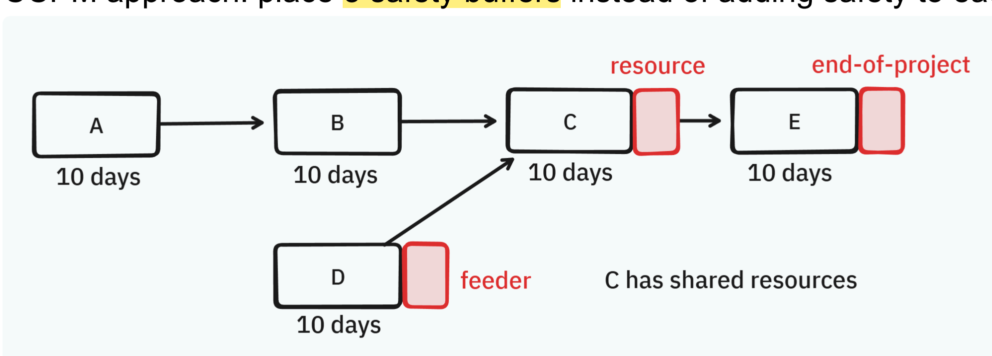


Solution II: Moving the schedule of one project



- splitting a task need to consider cost for Interruption of switching and Resuming a task

- Critical Chain Project Mgt. (CCPM)
  - people often inflate their estimateion since
    - your work depends on someone unreliable
    - past bad experience that things did not go as planned
    - the worry of project change
  - but project still late since
    - Student's Syndrome: 拖延症
    - Parkinson's Law: 如果提早匯報完成，下次能運用的時間可能會減少
    - Multitasking / Resource Contention: 通常一個人不會只參與一個 task
  - CCPM approach: place 3 safety buffers instead of adding safety to each task



    - Feeding buffer: add when a task is a feeder to tasks on critical path
    - Resource buffer: add when a task has shared resources
    - End-of-project buffer
- Schedule Compression
  - Crashing (趕工)

        - increase cost to decrease the duration of tasks
    - Fast Tracking (同步進行)
        - parallel performaing the ordered tasks
        - no dependency between parallel tasks
        - only useful for tasks on critical path

Project Cost Mgt.

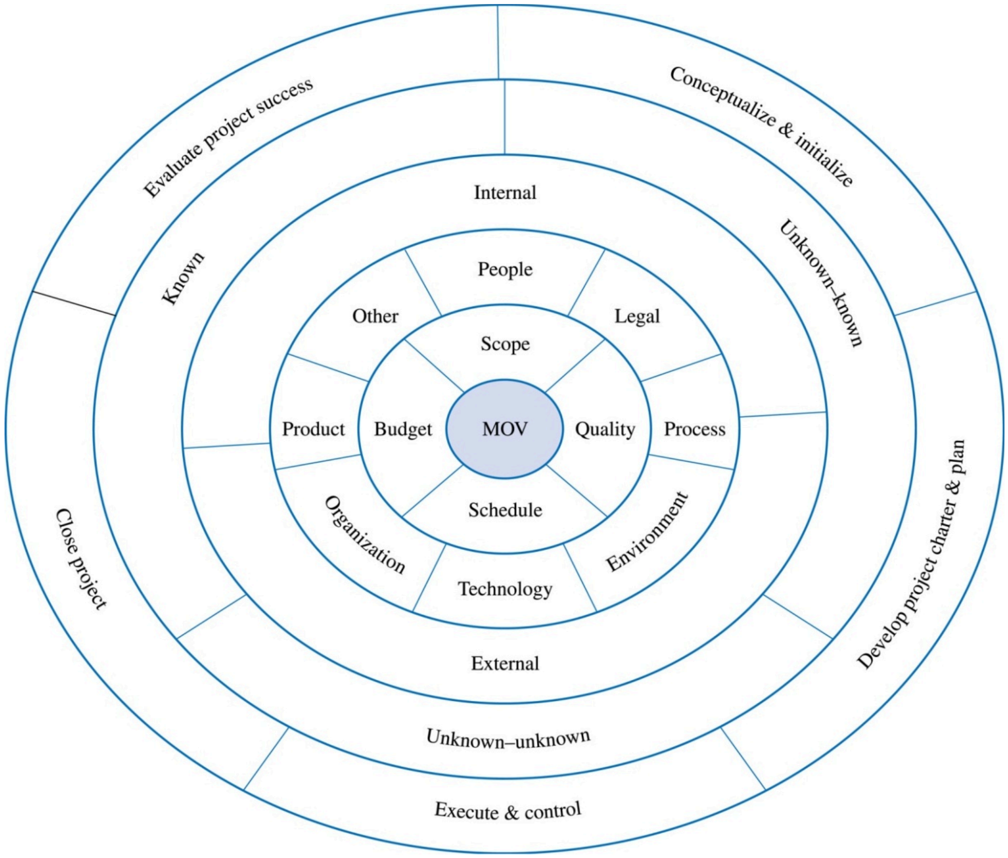
- PM approach:
    1. Compute & Sum up direct costs of each tasks in WBS
    2. Add other costs for the project
        - Indirect costs: 日常營運支出，通常用 coefficient 來當參考
        - Sunk costs: 已發生的成本 (e.g. cost of previous failed project)
        - Learning curve
        - Reserves: 用於突發事件的預備金，無使用(風險管理得當)則成為獲利
- SE approach
    - non-coding work can be estimate by the PM approach
    - to estimate software program:
        - Lines of Code
        - Function Points
            - estimate based on functionality defined by users
            - available in the earlier stage
            - 5 function types in 2 categories
                - Data Function Type
                    - ILF: Internal Logical File (內部資料庫) (讀取不算)
                    - EIF: External Interface File (外部資料庫)
                - Transaction Function Type
                    - EI: External Input
                    - EO: External Output
                    - EQ: External Inquiry
            - Early-phase FP:
                1. Obtain context diagram
                2. Indentify the 5 function types and weight by complexity level (given by company)
                    - $\mathrm{UFP} = \sum \text{weighted FP}$
                3. Consider adjustment factors by answering the 14 questions
                    - $\mathrm{VAF} = 0.65 + 0.01 \times \sum_{i=1}^{14} X_i$

4. Calculate the adjusted FP
    - $\mathrm{AFP} = \mathrm{UFP} \times \mathrm{VAF}$
- Design-phase FP:
  - As layout become clearer, we can obtain following information to better determine the complexity:
    - RET: Record Element Type (sub-tables)
    - DET: Data Element Type (data field)
    - FTR: File Type Reference (# of tables = ILF + EIF)
- COCOMO (COnstructive Cost Model)

1. Type	Organic	Semi-Detached	Embedded
2. $\mathrm{Effort} = M \times \mathrm{KLOC}^N$	$M = 2.4, N = 1.05$	$M = 3, N = 1.12$	$M = 3.6, N = 1.2$
3. $\mathrm{Duration} = 2.5 \times \mathrm{Effort}^N$	$N = 0.38$	$N = 0.35$	$N = 0.32$
4. $\mathrm{People} = \mathrm{Effort}/\mathrm{Duration}$			

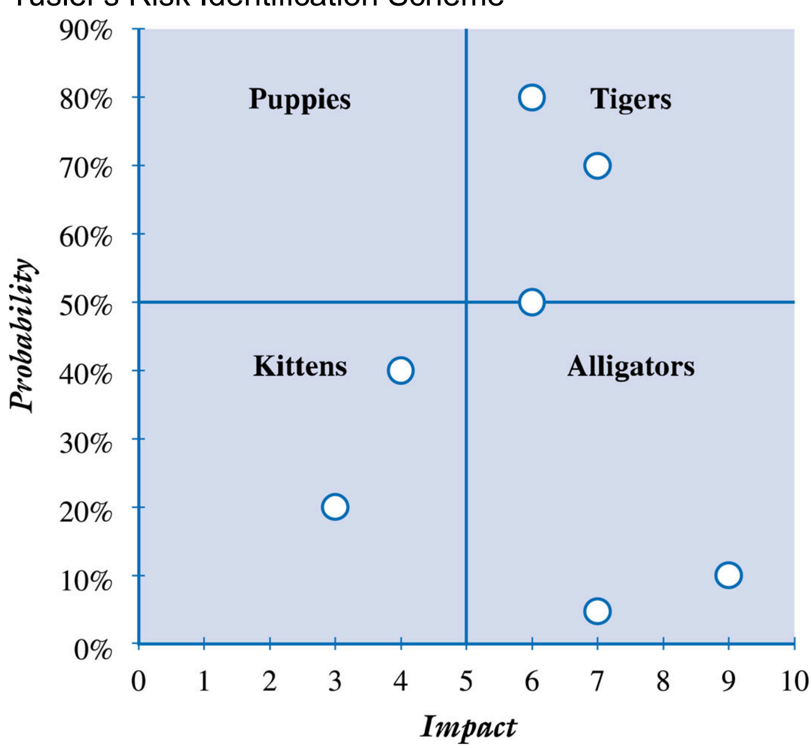
Project Risk Mgt.

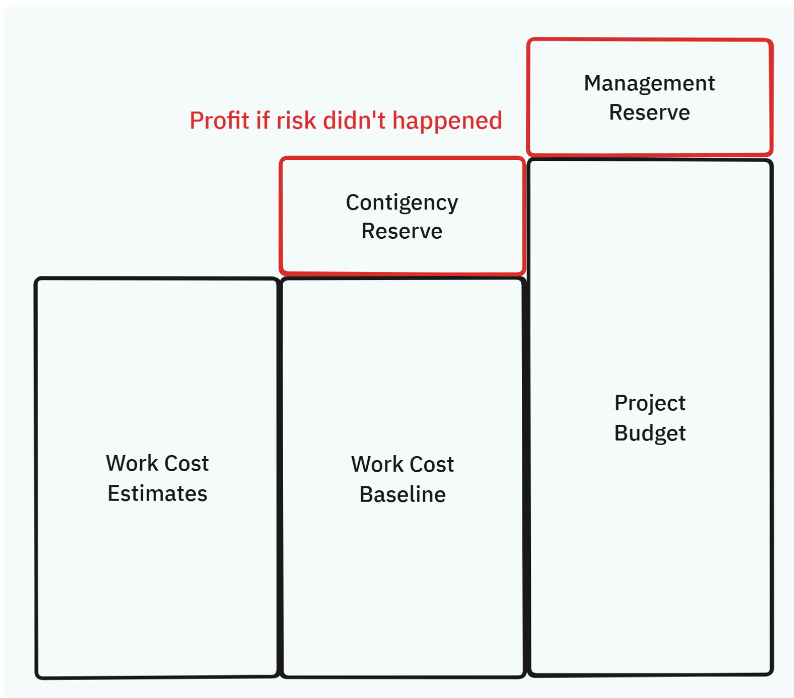
- Definition:
  - *Risk*: the effect of uncertainty on objectives, includes both negative & positive impacts
  - If not well handle: *Risk* → *Crisis* (loss caused by risks) → *Disaster*
  - *Residual Risk*: same risk remains after all response implemented
  - *Secondary Risk*: side effects of implementing a response
- Risk Management:
  1. Create a Risk Plan
  2. Indentify Risk
     - common risks: Technical, Managerial, Organizational, Customer, Supplier
     - Systemic Framework:
       從中心 (MOV) 開始，依序向外檢查每一層是否有危害到上一層的因素



- Risk Register: register identified risks and assign risk owner

3. Analyze Risks
    - Qualitative risk assessment: $\text{Risk Exposure} = \text{Probability} \times \text{Impact}$
        - Decision Tree Analysis
        - Risk Impact Table
        - Tusler's Risk Identification Scheme

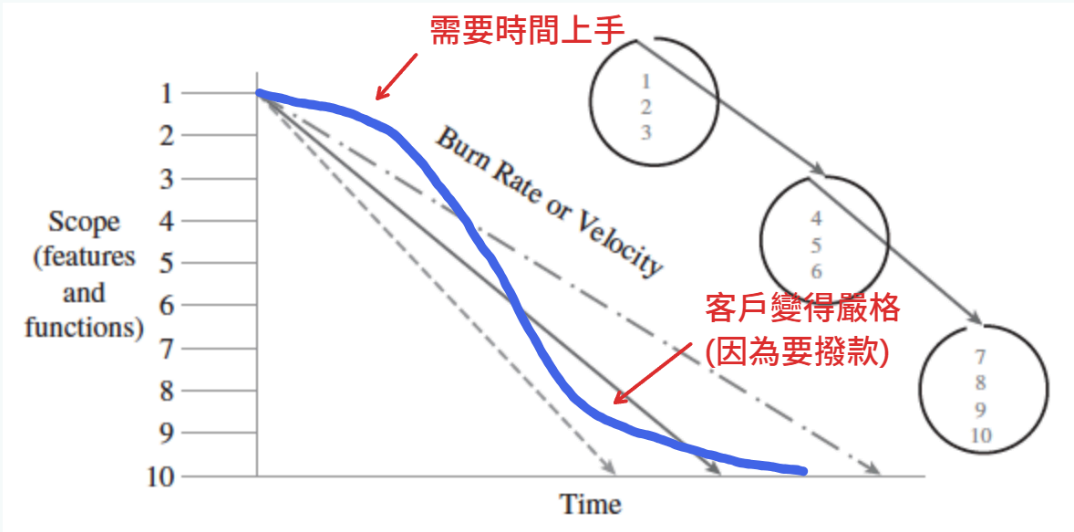


        - Probability/Impact Risk Rating Matrix
    - Quantiitative: observe the likeliness of a risk by probability distribution
4. Develop Risk Strategies
    - Strategies:
        - Accept/Ignore: for low prob. and low impact
        - Avoidance: eliminate the possibility
        - Mitigate: to reduce the likelihood or impact
        - Transfer: transfer risk ownership, e.g. buying insurance
        - Diversify: spreading risk out to limit negative effects on us
        - Escalate: reporting to higher level for help
    - Reserve:
        - 已知風險 → Contingency Reserve (CR): 風險發生時由 PM 支用
        - 未知風險 → Management Reserve (MR): 全公司共用，發生時由高層動用
        - 

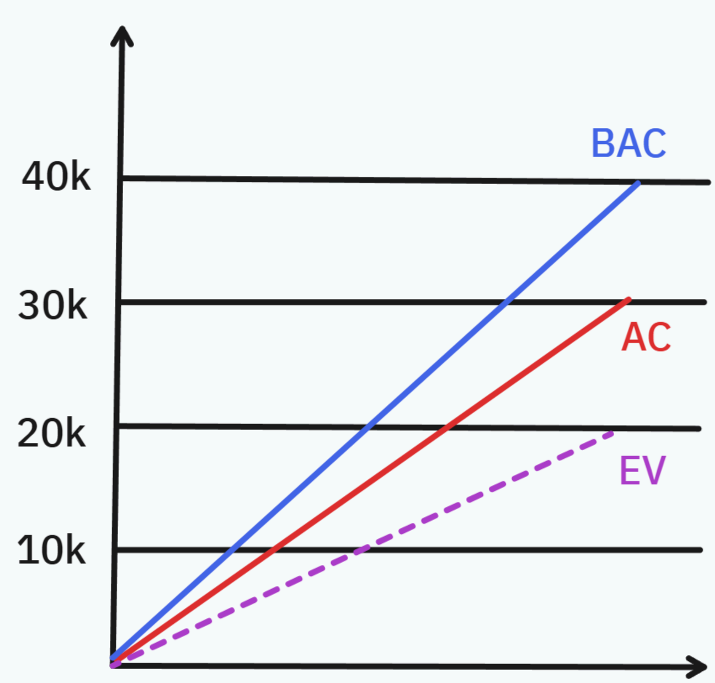
5. Monitor and Control
6. Respond and Evaluate

Project Progress Mgt.

- Burn-down chart: managing the progress to implement the scope



- Earned Value Method
  - Parameters:
    - Planned Value (PV): the budget **we planned to pay** for an task
    - Budgeted At Completion (BAC): total budget for project
    - Actual Cost (AC): actual cost **we have to pay** for completing an task
    - Earned Value (EV): the budget **we should pay** for the completed work
  - We expect to complete 4 tasks (10k each) in this month. (BAC = 40k)
    However only 2 task complete in the end (EV = 20k), and each task cost 15k. (AC = 30k)

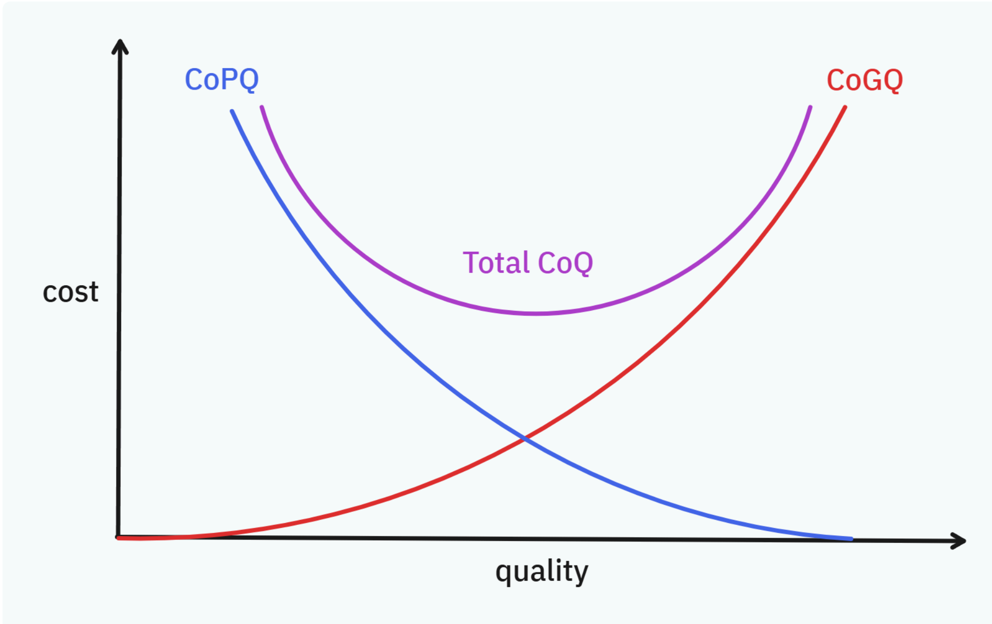


- Cost Metrics
    - Cost Variance: $\mathrm{CV} = \mathrm{EV} - \mathrm{AC}$
    - Cost Performance Index: $\mathrm{CPI} = \mathrm{EV}/\mathrm{AC}$
- Schedule Metrics
    - Schedule Variance: $\mathrm{SV} = \mathrm{EV} - \mathrm{PV}$
    - Schedule Performance Index: $\mathrm{SPI} = \mathrm{EV}/\mathrm{PV}$
- forecasting: Estimate at Completion (EAC)
    - typical (we believe the variances WILL continue): $\mathrm{EAC} = \mathrm{Cumulative\ AC} + (\mathrm{BAC} - \mathrm{Cumulative\ EV})/\mathrm{Cumulative\ CPI}$
    - atypical (we believe the variances WILL NOT continue): $\mathrm{EAC} = \mathrm{Cumulative\ AC} + (\mathrm{BAC} - \mathrm{Cumulative\ EV})$

- Managing by Issue
  - Anomalies (異常): cost, time , scope related
  - Defects (缺失): quality related
    - 沒有 follow SOP 也算缺失

Project Quality Mgt.

- the 3 types of requirements:
  - Project: budget, time, resources, people
  - Product: include software and non-software deliverables
  - Process: how the products are made
- Costs of Quality (CoQ)
  - CoPQ: cost of poor qulity
    - *ICF*: internal costs (e.g. 測試時發現的錯誤)
    - *ECF*: external costs (e.g. 使用者客訴)
  - CoGQ: cost of good quality
    - *AppC*: appraisal costs (e.g. 評鑑、認證)
    - *PrevC*: prevention costs (e.g. 設計驗證、開發測試)

- 

- 品管進化歷程: (quality by) Ispection → Manufactoring → Design → Management → Culture
- Quality Management Standards
  - For Product: 食品標章...
  - For Process: ISO, 6-Sigma, CMMI
  - CMMI (Capability Maturity Model Integration) v1.3
    - project's Maturity level is determined by process areas and their Capability levels
    - 22 *process areas*
      - each has *specific goals*, where each goal has *specific proctices*
      - develop *generic goals* and *generic practices* for sustainability
- Establish PQM Plan
  1. Establish Quality Policies
  2. Develop Metrics for selected Standard
     - consider the 3 types of requirements
  3. Plan Quality Assurance (QA)
     - V&V:
       - Verification (VER): Are we building the product in the right way?
         - focus on process
         - ensure the process are repeatable, the artifacts are of quality
       - Validation (VAL): Did we build the right product?
         - focus on product
         - ensure the system to serve user needs in intended environment
     - Methods of V&V:
       - **Review** (most VER):

        - Management review
        - Technical review
        - List-based inspection
        - Walk-through
        - Audit
    - **Test** (both):
        - unit testing
        - integration testing
        - systems testing
        - acceptance testing
        - regression testing
    - **Simulation** (most VAL)
    - **Demonstration** (most VAL)
    - **Pilot Try** (most VAL)

4. Plan Quality Control (QC)
    - monitor the quality of results, take corrective actions on the deviations
    - QA 和 QC 的區分需要看成果的定義
        - 在成果產出之前的預防 → QA
        - 在成果產出之後的檢查 → QC
5. Continual Improvement: PDCA, IDEAL